Multimedia Databases Exercises
SS 2026

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Exercise 1

Subject: Multimedia and Multimedia Databases - Initial Concepts
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Task 1: Signal Processing

a) In the context of the Pulse Code Modulation process, explain the sampling and quantization steps.

b) The basic formulation of a sine wave is given by $f(x) = A \cdot \sin(2\pi f x + \varphi)$. Given A = 10V, what is the maximum quantization error if a uniform quantization to 5 bits is applied to this wave?

c) State the Nyquist–Shannon sampling theorem. Based on this theorem, for the composite signal given in equation (1), calculate the minimum sampling rate so that the underlying information is not lost.

d) What is meant by the term aliasing?

$$f(x) = \sin(0.8\pi x) + \sin(1.5\pi x) + \sin(3\pi x) \tag{1}$$

Task 2: Structured and Unstructured Data

a) What are the characteristics of structured, unstructured and semi-structured data. Give examples for each.

b) What are the effects of such data on databases and their query properties?

Task 3: Application History - Compact Discs

Multimedia industries and standards evolved side by side with advancements in technology. One good example that illustrates this evolution are the ”Rainbow Books” that define Compact Disc standardizations.

a) Research the functional definitions and additions of each of the books

b) Apply the multimedia definitions and concepts from the lecture to your findings